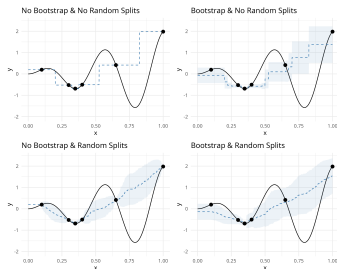
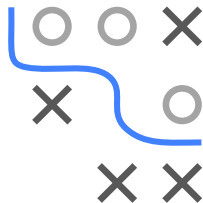


# Optimization in Machine Learning

## Bayesian Optimization Important Surrogate Models



### Learning goals

- Search space / input data peculiarities in black box problems
- Gaussian process
- Random forest

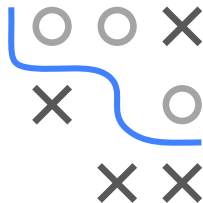
# SURROGATE MODELS

Desiderata:

- Regression model (there are also classification approaches)
- Non-linear local model
- Accurate predictions (especially for small sample sizes)
- Often: uncertainty estimates
- Robust, works often well without human modeler intervention

Depending on the application:

- Can handle different types of inputs (numerical and categorical)
- Can handle dependencies (i.e., hierarchical input)



# GAUSSIAN PROCESS

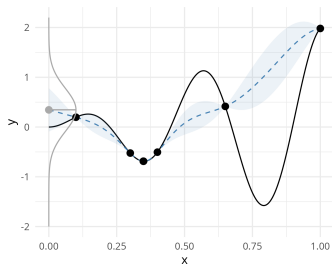
Posterior predictive distribution for test point  $\mathbf{x} \in \mathcal{S}$  under zero mean:

$$Y(\mathbf{x}) \mid \mathbf{x}, \mathcal{D}^{[t]} \sim \mathcal{N}(\hat{f}(\mathbf{x}), \hat{s}^2(\mathbf{x}))$$

with

$$\hat{f}(\mathbf{x}) = k(\mathbf{x})^\top \mathbf{K}^{-1} \mathbf{y}$$

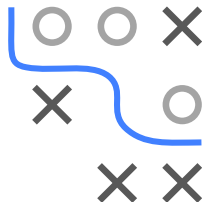
$$\hat{s}^2(\mathbf{x}) = k(\mathbf{x}, \mathbf{x}) - k(\mathbf{x})^\top \mathbf{K}^{-1} k(\mathbf{x})$$



Note:  $\mathbf{x}$  here denotes the test input.  $k(\mathbf{x}) := (k(\mathbf{x}, \mathbf{x}^{[1]}), \dots, k(\mathbf{x}, \mathbf{x}^{[t]}))^\top$ .

$\mathbf{y} := (y^{[1]}, \dots, y^{[t]})^\top$ .

Kernel / Gram matrix  $\mathbf{K} := \left( k(\mathbf{x}^{[i]}, \mathbf{x}^{[j]}) \right)_{i,j}$  where  $i, j \in \{1, \dots, t\}$ .



# GAUSSIAN PROCESS

Example kernel functions:

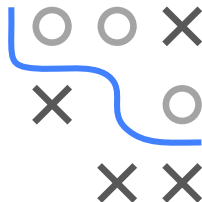
- Radial basis function kernel (also known as Gauss kernel):

$$k(\mathbf{x}, \mathbf{x}') = \exp\left(-\frac{d(\mathbf{x}, \mathbf{x}')^2}{2l^2}\right)$$

- $l$  length scale;  $d(\cdot, \cdot)$  Euclidean distance
- infinitely differentiable - very “smooth”

- Matérn kernels:  $k(\mathbf{x}, \mathbf{x}') = \frac{1}{\Gamma(\nu)2^{\nu-1}} \left(\frac{\sqrt{2\nu}}{l} d(\mathbf{x}, \mathbf{x}')\right)^\nu K_\nu\left(\frac{\sqrt{2\nu}}{l} d(\mathbf{x}, \mathbf{x}')\right)$

- $l$  length scale;  $d(\cdot, \cdot)$  Euclidean distance;  $K_\nu(\cdot)$  modified Bessel function;  $\Gamma(\cdot)$  Gamma function
- for  $\nu = 3/2$  once differentiable, for  $\nu = 5/2$  twice differentiable
- Popular choice as a kernel function when using a GP as SM



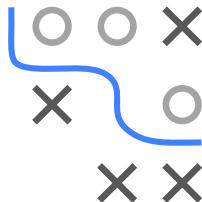
# GAUSSIAN PROCESS

## Pros:

- Smooth, local, powerful estimator, also for small data
- GPs yield well-calibrated uncertainty estimates
- The posterior predictive distribution under a GP is normal

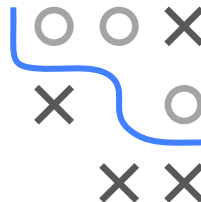
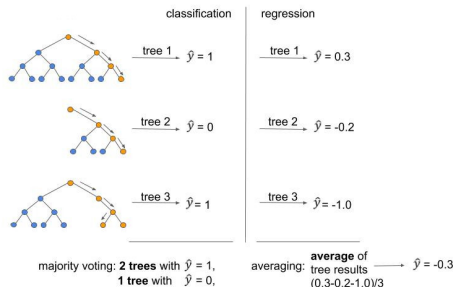
## Cons:

- Vanilla GPs scale cubic in the number of data points
- Can natively only handle numeric features Mixed inputs / dependencies require special kernels
- GPs aren't that robust; numerical problems can occur
- Can be sensitive to the choice of kernel and hyperparameters



# RANDOM FOREST

- Bagging ensemble
- Fit  $B$  decision trees on bootstrap samples
- Feature subsampling



“extratrees” / random splits:

- Choose split location uniformly at random
- Results in a “smoother” mean prediction

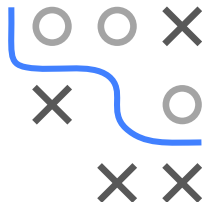
# RANDOM FOREST - MEAN AND VARIANCE

- Let  $\hat{f}_b : \mathcal{S} \rightarrow \mathbb{R}$  be the mean prediction of a decision tree  $b$  (mean of all data points in the same node as observation  $\mathbf{x} \in \mathcal{S}$ )
- Let  $\hat{s}_b^2 : \mathcal{S} \rightarrow \mathbb{R}$  be the variance prediction (variance of all data points in the same node as observation  $\mathbf{x} \in \mathcal{S}$ )
- Mean prediction of forest:  $\hat{f} : \mathcal{S} \rightarrow \mathbb{R}, \mathbf{x} \mapsto \frac{1}{B} \sum_{b=1}^B \hat{f}_b(\mathbf{x})$
- Variance prediction of forest:  $\hat{s}^2 : \mathcal{S} \rightarrow \mathbb{R},$

$$\mathbf{x} \mapsto \left( \frac{1}{B} \sum_{b=1}^B \hat{s}_b^2(\mathbf{x}) + \hat{f}_b(\mathbf{x})^2 \right) - \hat{f}(\mathbf{x})^2$$

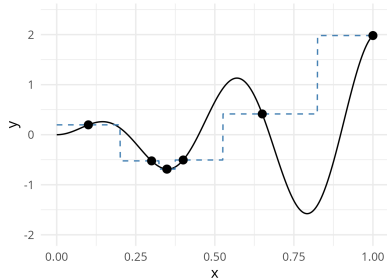
(law of total variance assuming a mixture of  $B$  models)

- Alternative variance estimator:
  - (infinitesimal) Jackknife
- Variance prediction derived from randomness of individual trees
  - Bagging / bootstrap samples
  - Features sampled at random
  - (randomized split locations in the case of “extratrees”)

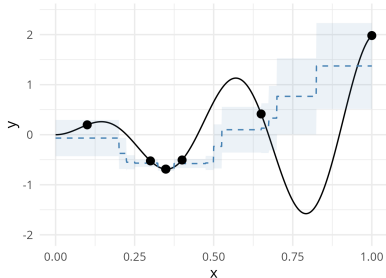


# RANDOM FOREST - DIFFERENT CHOICES

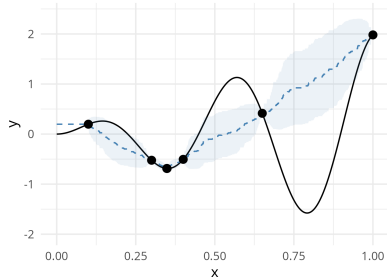
No Bootstrap & No Random Splits



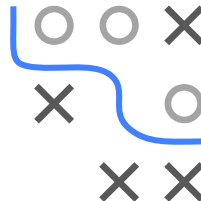
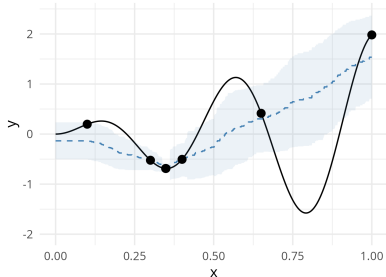
Bootstrap & No Random Splits



No Bootstrap & Random Splits



Bootstrap & Random Splits





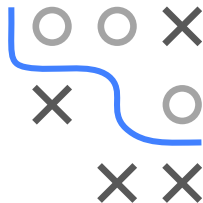
# RANDOM FOREST

## Pros:

- Cheap(er) to train
- Scales well with the number of data points
- Scales well with the number of dimensions
- Can easily handle hierarchical mixed spaces. Either via imputation or directly respecting dependencies in the tree structure
- Robust

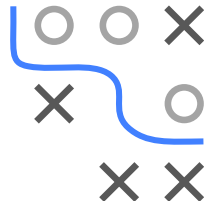
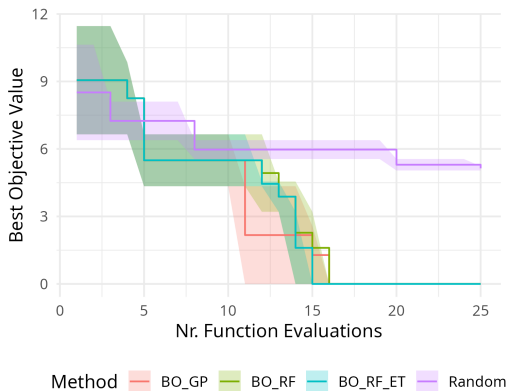
## Cons:

- Suboptimal uncertainty estimates
- Not really Bayesian (no real posterior predictive distribution)
- Poor extrapolation



# EXAMPLE

Minimize the 2D Ackley Function using BO\_GP (GP with Matérn 3/2, EI), BO\_RF (standard Random Forest, EI), BO\_RF\_ET (Random Forest with extratrees, EI) or a random search:



Strong BO\_GP performance. BO\_RF and BO\_RF\_ET not too bad either. BO\_RF\_ET maybe slightly better final performance than BO\_RF.